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5 Conclusion générale

— Chapitre 01 : L'organisme d'accueil

In this chapter, we present the host organization, its main activities, and the services it provides.

— Chapter 02: Existing System Study

This chapter is dedicated to analyzing the current management methods within the daycare, identifying the limitations of the existing system, and defining the problem as well as the objectives of our solution.

— Chapter 03: Analysis and Design

This chapter presents the development environment, the application architecture, the tools used, and the various implemented features

— Chapter 04: Implementation

Ce chapitre présente l'environnement de développement, l'architecture de l'application, les outils utilisés ainsi que les différentes fonctionnalités implémentées.

— General Conclusion

Finally, we conclude with a general summary of the results obtained and propose perspectives for future improvements.

Chaptre 01

General Context of The Project

Introduction

Daycare centers are really important because they take care of children. They help children learn and grow. Daycare centers do a lot of things for kids.

They teach children, Help them develop. Even though daycare centers are important, a lot of them still do things the old way. This is especially true in Algeria. Most daycare centers use paper or simple things like Excel files to manage things.

Using paper and simple tools causes a lot of problems. For example, daycare centers can lose information easily. Information gets scattered over the place. They have trouble keeping track of everything. They make mistakes when tracking the children. They get confused about who's present who paid and who is on staff.

Doing things by hand takes a time. It is repetitive. Sometimes it is not even reliable. This can affect how smoothly the daycare center runs. Because of these problems, daycare centers need to start using computers and technology. Using an application would make it easier to keep track of information. It would do some tasks automatically. It would help everyone work together better.

Our project is trying to help daycare centers. We want to make a computer program, for daycare centers. This program will help them manage things in an simple way. We want to help daycare centers use computers and technology to make things better.

Problem Statement

In spite of the significance of these facilities, there are some issues related to the management activities that the daycares have to address because of the inefficiency of the existing system. There are certain problems concerning the way the information is collected and managed, and they cause challenges for effective management.

These issues make the work of administrators complicated and time-consuming, creating the threat of making mistakes while handling important processes, including attendance registration, fee collection, record presence, etc. They complicate communication and organization of the process.

It should be noted that the increase in the number of children means increasing the workload of the administration and complicating their work further.

It proves to be vital for the management to create a more efficient system for collecting and managing the information.

Proposed Solution

In order to solve all the issues raised in the management of daycares, it is necessary to implement a desktop application that will organize all aspects of the process in one system and allow easier and faster access to any kind of data.

As a result of implementing this solution, it will be possible to perform such functions as managing the records of children, managing their presence, managing the information of personnel and payments more effectively and accurately.

Furthermore, such useful functions as searching, sorting, and updating will facilitate the management process significantly.

Besides, the use of statistical instruments will allow the management of a daycare to analyze its operation and performance better.

All in all, the implementation of this solution will lead to higher efficiency, accuracy, better organization, as well as more modern communication.

Conclusion

In this chapter, we have introduced the general scenario in which our system will work in, emphasizing the significance of daycare centers along with their role in helping children develop and assist families. In addition to this, we explained the current situation of these institutions wherein most of them use conventional and manual systems for handling their routine activities.

Following the discussion regarding these shortcomings, we determined some important challenges that make the day-to-day operation at these places difficult. It is clear from these limitations that there is a need for adopting a more advanced approach in order to ensure the improvement in their operations.

Ultimately, we introduced a computerized solution in the form of desktop software that would be aimed at centralizing information and improving organizational performance. The proposed solution will form the foundation for our system as we continue to discuss analysis, design, and implementation.

Chaptre 02

Existing System Study

Introduction

Being an important component of developing any computer application, one should start with the analysis of the present situation, as it provides an opportunity to gain insight into how things work within the given field at the moment.

The current reality shows that the administrative management in day-care centers is usually implemented using conventional approaches, such as writing notes on paper or working manually. Such a practice may cause different organizational and technical issues.

Thus, this chapter will focus on the analysis of the process of administrative management in day-care centers, discussing the shortcomings of the current system as well as formulating a problem statement and objectives for implementing the project.

Analysis of Current Processes

General Overview: The process involved in the running of childcare centers is normally done by traditional means whereby there is involvement of manual processes using simple tools like paper and excel files. This is done in order to manage the various operations in the day care center such as enrollment of children and payments.

Information Handling: In most cases, information is recorded manually and stored in different documents, which makes it difficult to organize and update data efficiently. Each process is handled separately, and there is no centralized system to manage all the information in one place.

Operational Difficulties: As such, the employees will have to allocate significant time in finding, updating, and confirming the information in the system. This adds up to their tasks and can result in delays and mistakes in operations.

Conclusion of the Analysis: Therefore, analyzing these current processes helps to better understand the weaknesses of the existing system and highlights the need for an improved and automated solution.

Children Management

Generally, the management and storage of information on children in these institutions takes place manually by means of paper records (registers, etc.) or simply computerized documents. Each child has their record where basic information like name, age, place of residence, and data of contacts (parents) are stored. They are mostly kept in paper folders or other media, which complicates their organization and maintenance.

In case any changes need to be introduced into this document (for example, updating data about the person), then they have to be found manually by staff, which takes much time and effort and may result in making mistakes.

Moreover, monitoring and recording any history regarding each kid, like attendance or payments, is poorly organized.

August 28, 2023

[Click Here to Return to First Record in Enrollment List](#)

Example: 6 months old = 1 year old.
Example: 3 years, 6 Months = 4 years old

Child's Name	Enrollment Status	Enrollment Date	Child's Birthday	Classroom	Parent/Guardian Name	Primary Contact Number	eMail Address
Will Anderson	Active	5-Sep-23	18-Jan-19	Lions	Kristina Anderson	(406) 560-2222	Kristina.Anderson@gmail.com
Ella Smith	Active	5-Sep-22	5-Dec-19	Bears	Whitney Smith	(406) 222-1111	Whitney.Smith@gmail.com
Micah Williams	Active	5-Sep-22	4-Apr-20	Classroom H	Michelle Williams	(406) 523-4789	Micah.Williams@gmail.com
Jonah Price	Active	6-Aug-22	10-Jun-20	Classroom H	Tawny Price	(406) 789-4563	TanwyPrice@gmail.com
Krissy Stone	Active	5-Sep-22	26-Jun-21	Classroom G	Jen Stone	(406) 560-2258	JennyStone@gmail.com
Tommy Brown	Active	5-Sep-22	18-Jan-19	Tigers	Devin Brown	(406) 560-3146	DevinBrown@outlook.com
Cooper Lum	Active	5-Sep-23	18-Jan-19	Lions	Ashley Lum	(406) 555-7788	AshleyLum@gmail
Elsa Winter	Active	5-Sep-22	5-Dec-19	Lions	Corey Winter	(406) 222-1111	Corey.winter@gmail.com
Desiree Kringle	Active	5-Sep-22	4-Apr-19	Lions	Theresa Kringle	(406) 523-4789	Theresa.Kringle@gmail.com
Poppy Gibben	Active	6-Aug-22	25-Apr-21	Classroom K	Tawny Price	(406) 789-4563	TanwyPrice@gmail.com
Vivian Gibben	Active	5-Sep-22	25-Apr-21	Classroom K	Jen Stone	(406) 560-2258	JennyStone@gmail.com
Carson Anderson	Active	5-Sep-22	4-Jul-20	Classroom I	Kristina Anderson	(406) 560-1234	DevinBrown@outlook.com
Dutch Harmon	Active	5-Sep-22	4-Jul-20	Classroom I	Howard Harmon	(406) 560-9874	DevinBrown@outlook.com
Karley White	Active	5-Sep-22	4-Jul-20	Classroom I	Karen White	(406) 560-2222	DevinBrown@outlook.com
David Forester	Active	5-Sep-22	4-Jul-20	Classroom I	Kathy Forester	(40) 656-3143	DevinBrown@outlook.com

Figure 2.1: Example of children register used in daycare centers

Attendance Management

Management of attendance at daycare centers is typically performed manually through the use of paper registers and tables. Educators fill out specific forms for each day depending on whether the child attended or not.

This system usually involves the use of physical forms with data input done daily, making access and updating of the data quite challenging. When mistakes happen, it is difficult to modify the records correctly.

Moreover, monitoring attendance within a certain period of time would require the analysis of several different registers, making the whole process very lengthy and possibly confusing.

Childcare Schedule (Enter "X" for each day attending)						
MON	TUES	WED	THU	FRI	SAT	SUN
X	X	X	X	X		
X		X		X		
X	X	X	X	X		
X	X		X	X		
X		X		X		
X	X	X	X	X		
X	X	X	X	X		
X	X	X	X	X		
	X	X	X			
X	X	X	X	X		
X		X		X		
X	X	X	X	X		
X	X	X	X	X		
X	X	X	X	X		

Figure 2.2: Example of attendance table in Excel

Employee Management

Employee management in daycare centers is generally carried out in a traditional way using paper files. Each employee has a file containing personal information such as name, surname, contact details, job position, salary, and working schedule.

This information is often stored in a non-centralized manner, which makes it difficult to access and update. Any modification of data requires manual intervention, which increases the risk of errors or inconsistencies in the records.

In addition, tracking employees, such as managing attendance, absences, or schedules, is often done separately and without an automated tool. This can lead to a lack of organization and a loss of time in the daily management of staff.

Payment Management

Payments in childcare institutions are usually made using the conventional method by maintaining manual records and documents. The payments made manually are usually noted along with details such as the amount of money received, the month in which the payment is made, and the date of payment for each child.

Usually, payments are made by issuing a receipt for payment that is manually filled. Such a receipt serves as a proof of the payment that is made by parents.

This process of keeping track of payments is inefficient as the documents are maintained individually and can be difficult to track and monitor. Checking the history of previous payments or even the payment status of a particular child becomes very tedious.

Criticism of the Existing System

The current process for management in daycare centers relies mostly on manual processes and tools. Even though the process might still be functional, there are certain problems that make the process less effective.

For example, one of the most evident weaknesses of the current process is its unreliability. The fact that all data is manually collected and stored makes it vulnerable to inaccuracies and mistakes. For instance, the information about children, employees, attendance, and payments could be incorrect.

Secondly, there is a problem related to the way information is organized in the current process. The information is dispersed between different registers and files, which means that it is harder to locate and use.

Thirdly, the current process is non-automated and requires employees to do everything manually. It becomes a problem in terms of increasing efficiency because the employees have to work on multiple tasks without any assistance from computers.

Finally, the current process lacks digitalization and a central repository for storing information.

Objectives of the Solution

The main objective of the proposed solution is to improve the management of daycare centers by developing a centralized and automated system that replaces traditional manual methods. This system aims to simplify daily operations and make information handling more efficient and reliable.

One of the key objectives is to centralize all data related to children, employees, attendance, and payments in a single system, allowing easy access and better organization of information.

Another important objective is to reduce human errors by automating repetitive tasks such as recording attendance, managing payments, and updating records. This will improve the accuracy and reliability of the data.

The system also aims to save time and increase productivity by making administrative tasks faster and more efficient. Staff members will be able to focus more on essential tasks instead of manual data entry.

Finally, the solution aims to provide better monitoring and reporting tools, such as statistics and summaries, to help administrators make informed decisions and improve the overall management of the daycare center.

Daily Usage Flow Diagram

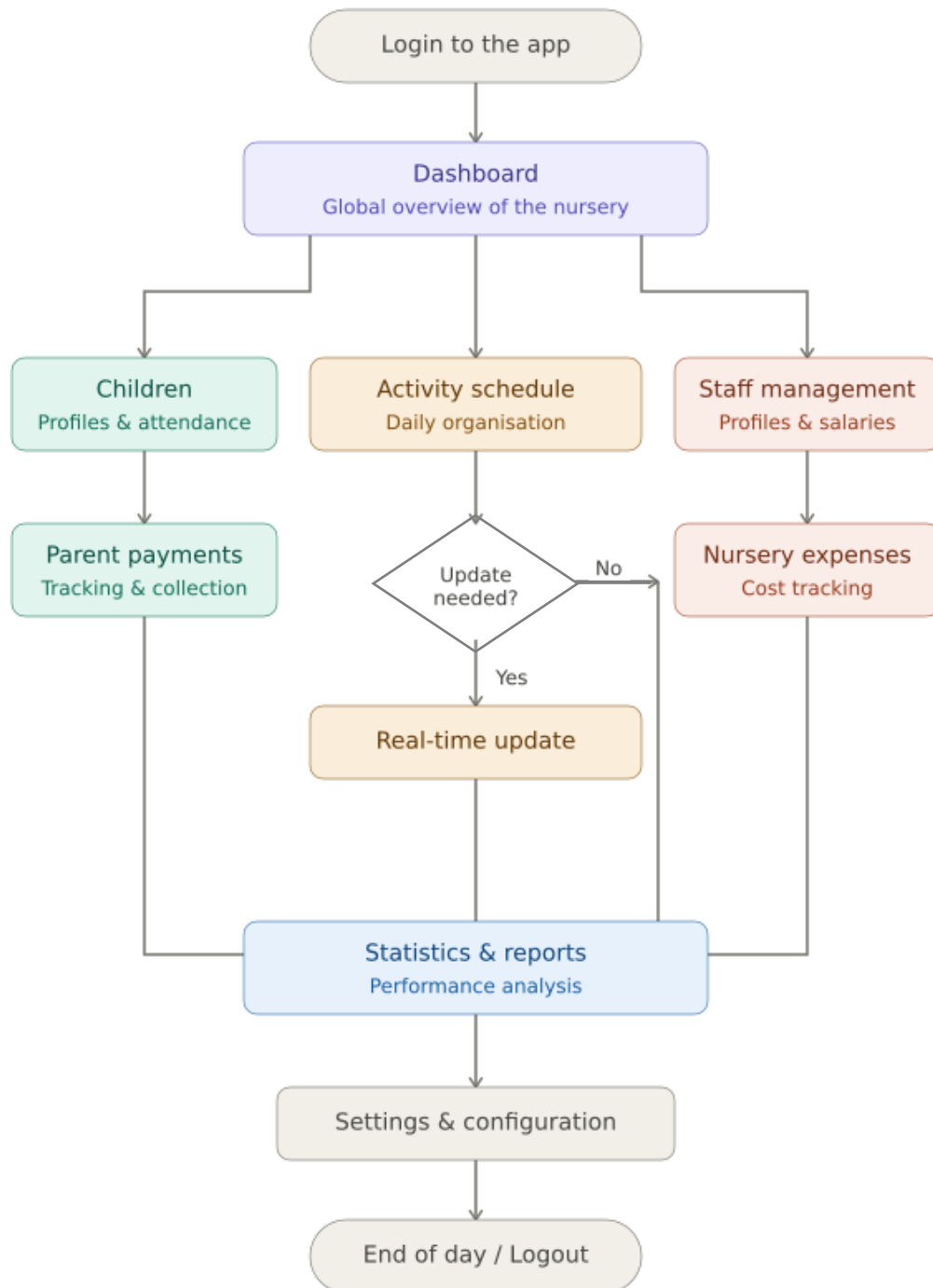


Figure 2.3: Software usage flow diagram in daily operations

Conclusion

In this chapter, we have done an analysis of the existing system used in daycare centers for carrying out day-to-day activities. We have considered processes such as processes relating to children, process relating to attendance, processes relating to employees, and processes relating to payments and realized that all these processes are executed in a manually-driven way using simple techniques.

The identified problems included non-centralized nature, chance of errors, time consumption, and difficulties in managing and updating information. These problems indicated clearly that the existing system does not cater to all the necessary requirements in order to execute its operation successfully.

Based on the above study, we have come up with major problems facing the system and its objectives. We have provided flowchart representation of the existing system and its usage.

Thus, from the above chapter, it has become evident that there exists a need for a more modern and centralized system and we shall describe it in subsequent chapters..

Chaptre 03

General Context of The Project

Introduction

After analyzing the current system and highlighting its weaknesses in the last chapter, the next step would be to concentrate on the analysis and design of the proposed system. The importance of this phase cannot be overstated when it comes to developing a product because it provides insight into the structure of the system and the interaction between its various components.

This phase is crucial since it allows us to grasp the requirements for the system before implementing it. It also helps us identify the actors, functionalities, and constraints that should be met to create a robust and efficient system.

In this chapter, we will develop the model using UML diagrams. We will also provide a brief overview of the system's functional and non-functional requirements, methodology used, and design.

UML Presentation

UML (Unified Modeling Language) refers to a standardized graphical language that can be used in software engineering for modeling, designing and documenting systems. UML offers a set of diagrams to effectively represent the structure and behavior of a system.

UML is a widely adopted standard in software development since it facilitates having a common understanding of the system prior to developing it. UML helps in representing system requirements, organizing concepts and identifying the relationships among different components within the system.

In this project, UML has been applied in modeling the daycare management system from various aspects. In our model, we mainly use the following diagrams; use case diagram, class diagram, sequence diagrams and database schema.

UML is used in our project to facilitate better organization of the system design.

Adopted Methodology

Waterfall methodology has been selected for the development of the given project as it is very simple and is commonly used in software engineering due to its phases being very clear.

Waterfall methodology consists of a sequence of steps like: requirements analysis, system design, implementation, testing, and maintenance. In order to perform the next step, each of these phases should be finished first.

In the case of our nursery management software, the usage of this methodology can be justified by such fact that the requirements of the system are very well defined initially and its development can be easily organized.

Thus, the first step of using waterfall methodology will be performing requirement analysis. Further, a design will be created according to the results of requirement analysis. As it was mentioned earlier, it will be created in the form of UML diagrams.

Further, after designing the system, its implementation and testing will take place.

Functional Analysis

Child Management:

This software helps users to add, update, view full profiles, and delete any information concerning the children. Every child has a comprehensive profile containing personal data, health information, information regarding their parents, the people authorized to pick them from school among others. Printing of the file and ID cards is also possible.

Attendance Management:

In order to facilitate attendance management, the program stores daily attendance of the children and creates an attendance history using the calendar format.

Employee Management:

To manage information on employees, the program uses the employee management option. This option helps in organizing and storing of information of the employees in the database.

Payment Management:

Payment management module is responsible for handling payments for each child. This module captures information related to payment period, payment made, mode of payment, and whether payment was made completely or not.

Statistics and Reports:

Lastly, the program comes with a statistical module where various statistics can be produced based on children, attendance and payments

Identification of Actors

In this system for nursery management, there will be users with different roles and responsibilities.

The primary actor in this system is the Administrator who will have complete access to the system. The administrator is responsible for managing the records of children, employees, attendance, payments, and analyzing statistics. Also, he can make changes to the data whenever necessary.

Secondly, another important actor is the Employee (Educator). This user will use the system to record daily attendance and analyze the information of children. They will have limited access to the system compared to the administrator.

A third optional actor can be Parent who is an external actor using the system indirectly. For example, they may need information related to their child like attendance and payment.

The above actors will determine the different levels of interaction with the system.

UML DIAGRAMS

Global Use Case Diagram

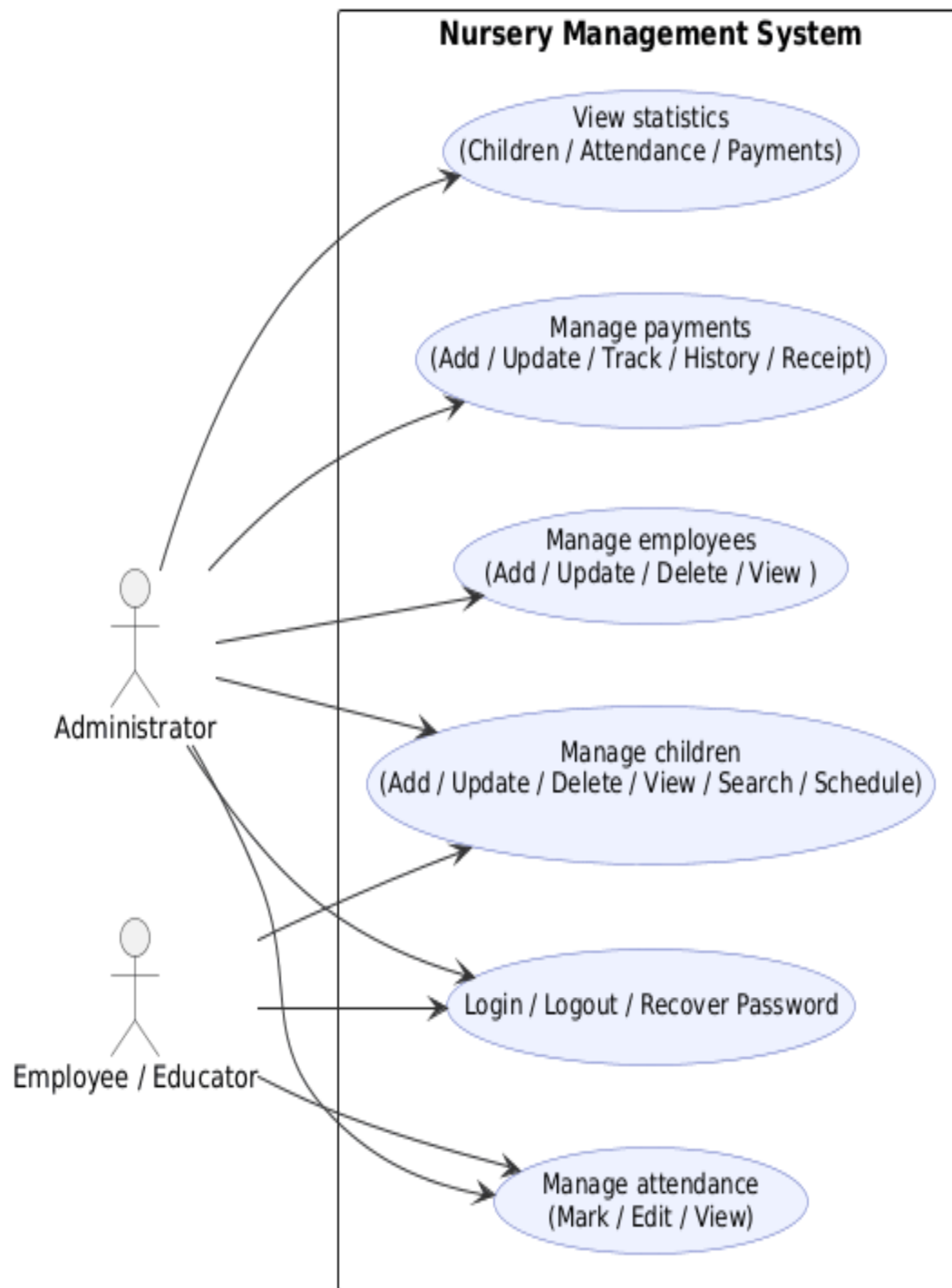


Figure 2.2: Global Use Case Diagram

Class Diagram

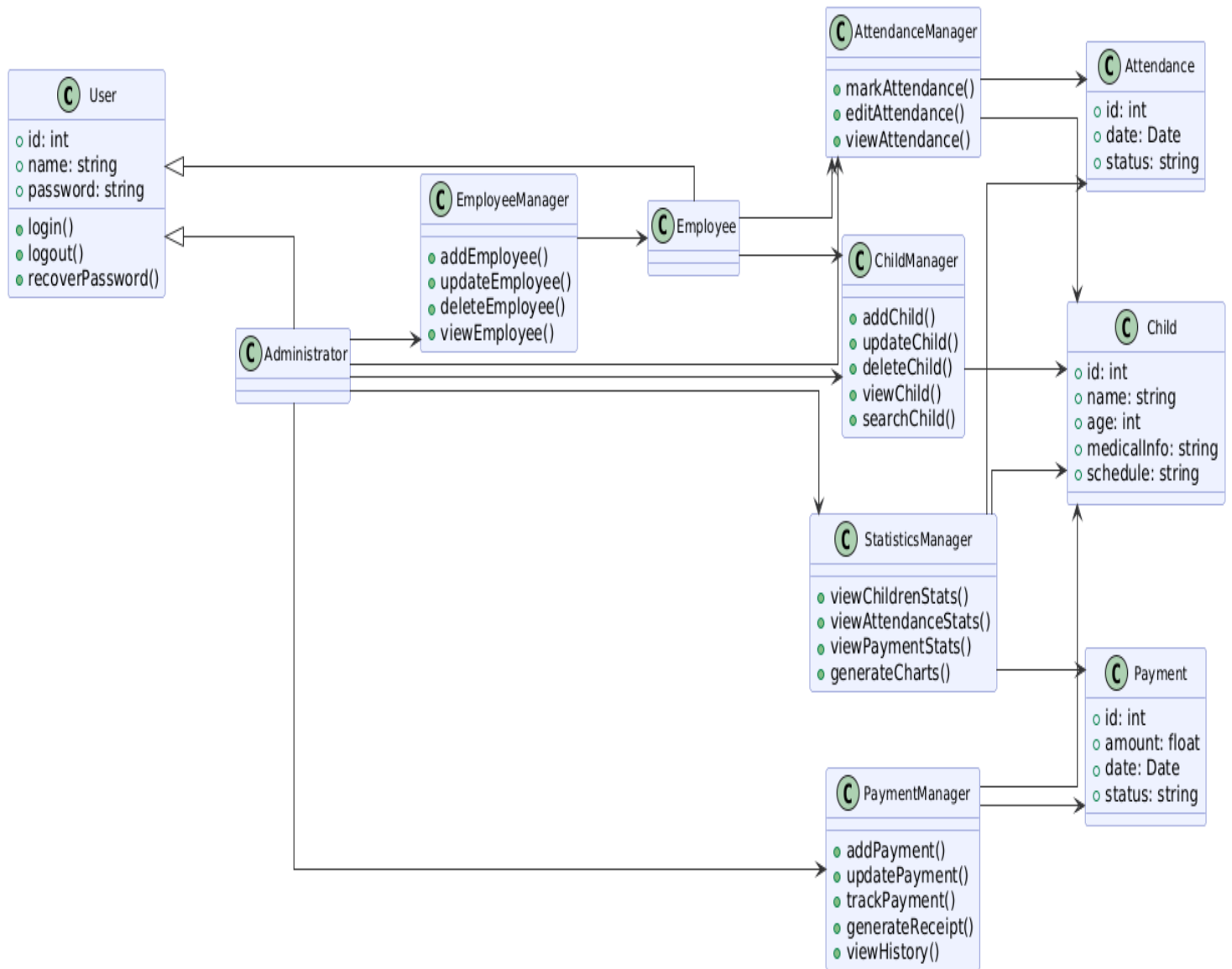


Figure 2.2: Class Diagram

Database Schema

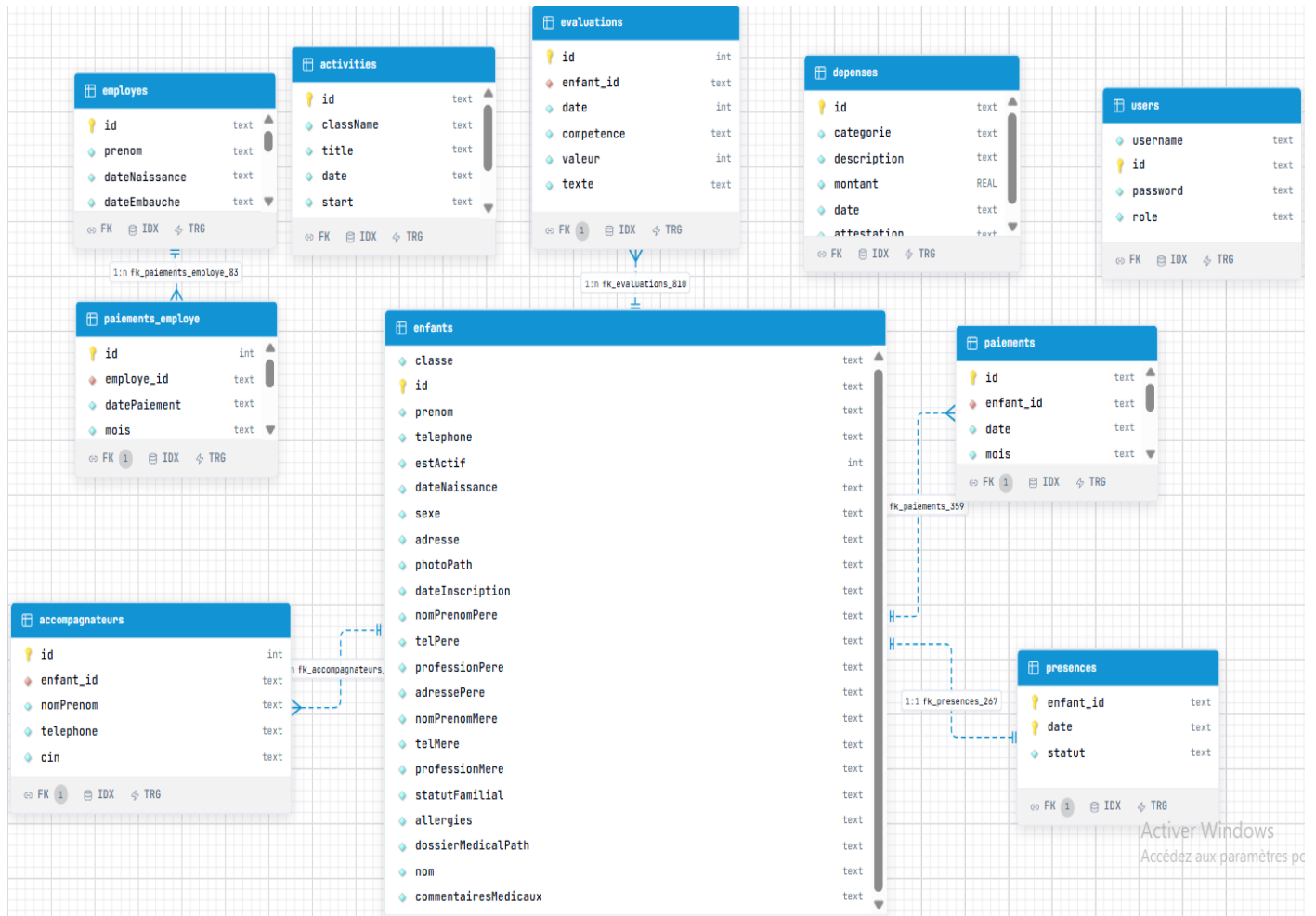


Figure 2.2: Database Schema

Sequence Diagram

Authenticcation:

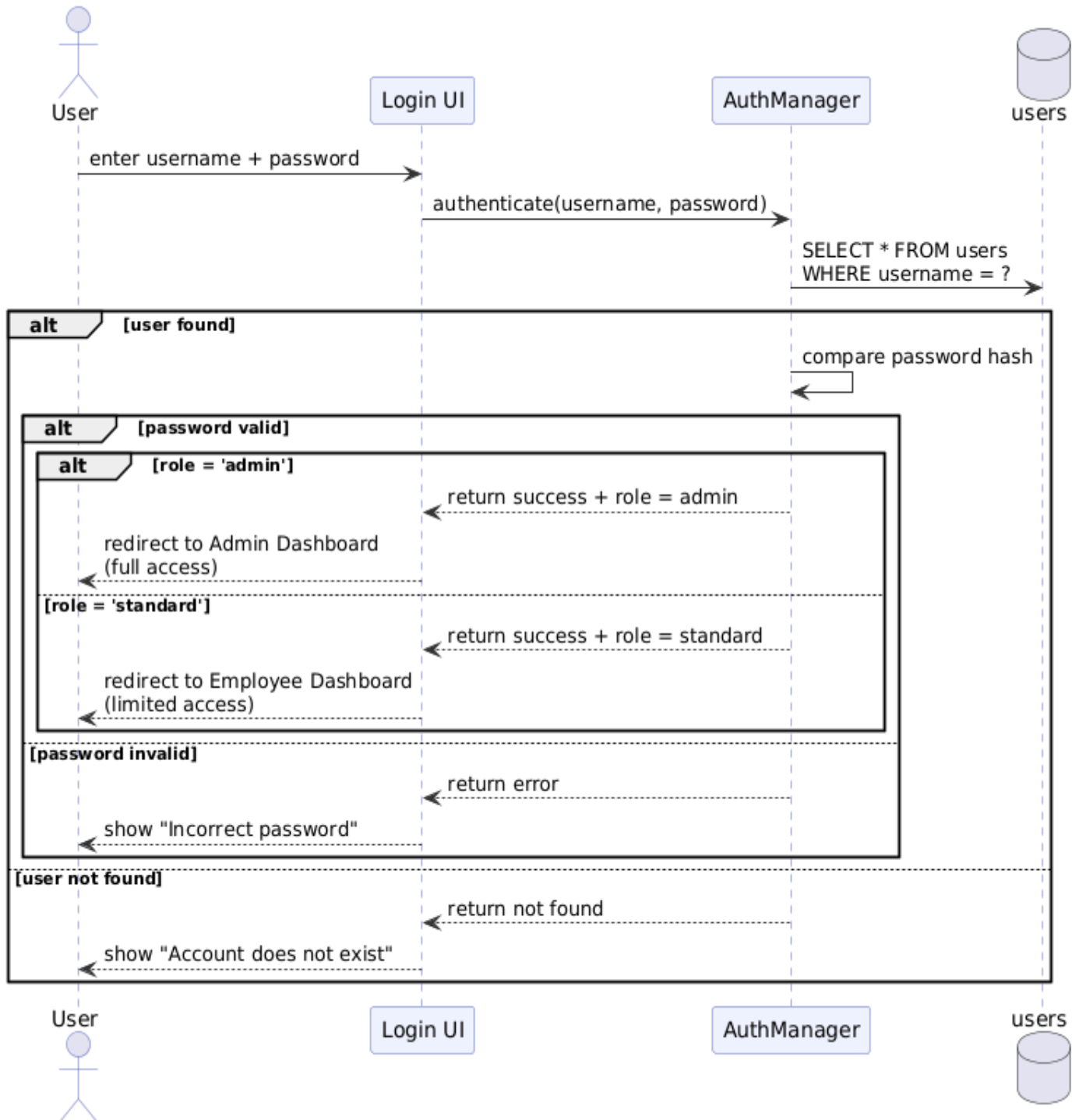


Figure 2.2: Authentication Sequence Diagram

Child Managment:

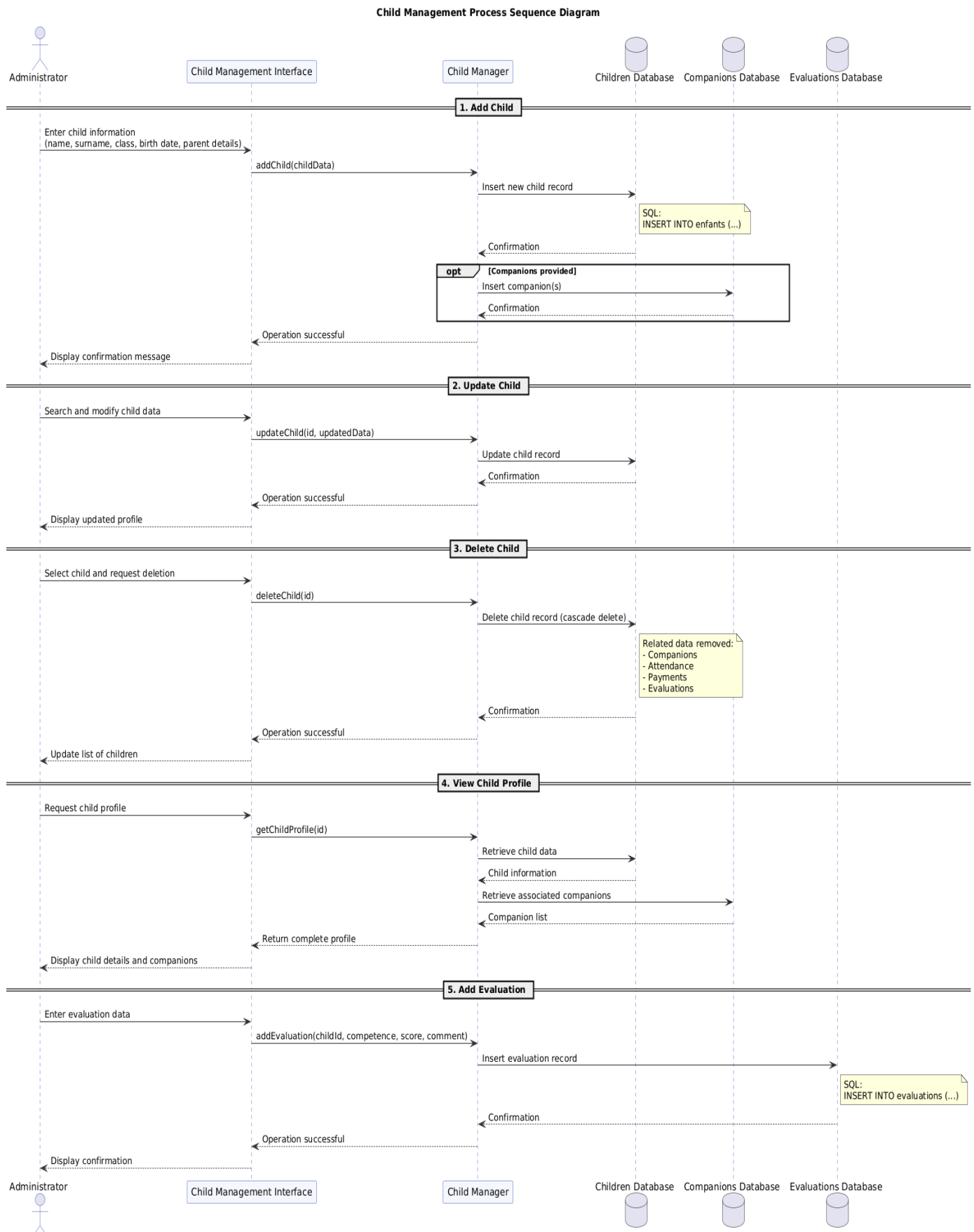


Figure 2.2: Authentication Sequence Diagram

Attendance Managment:

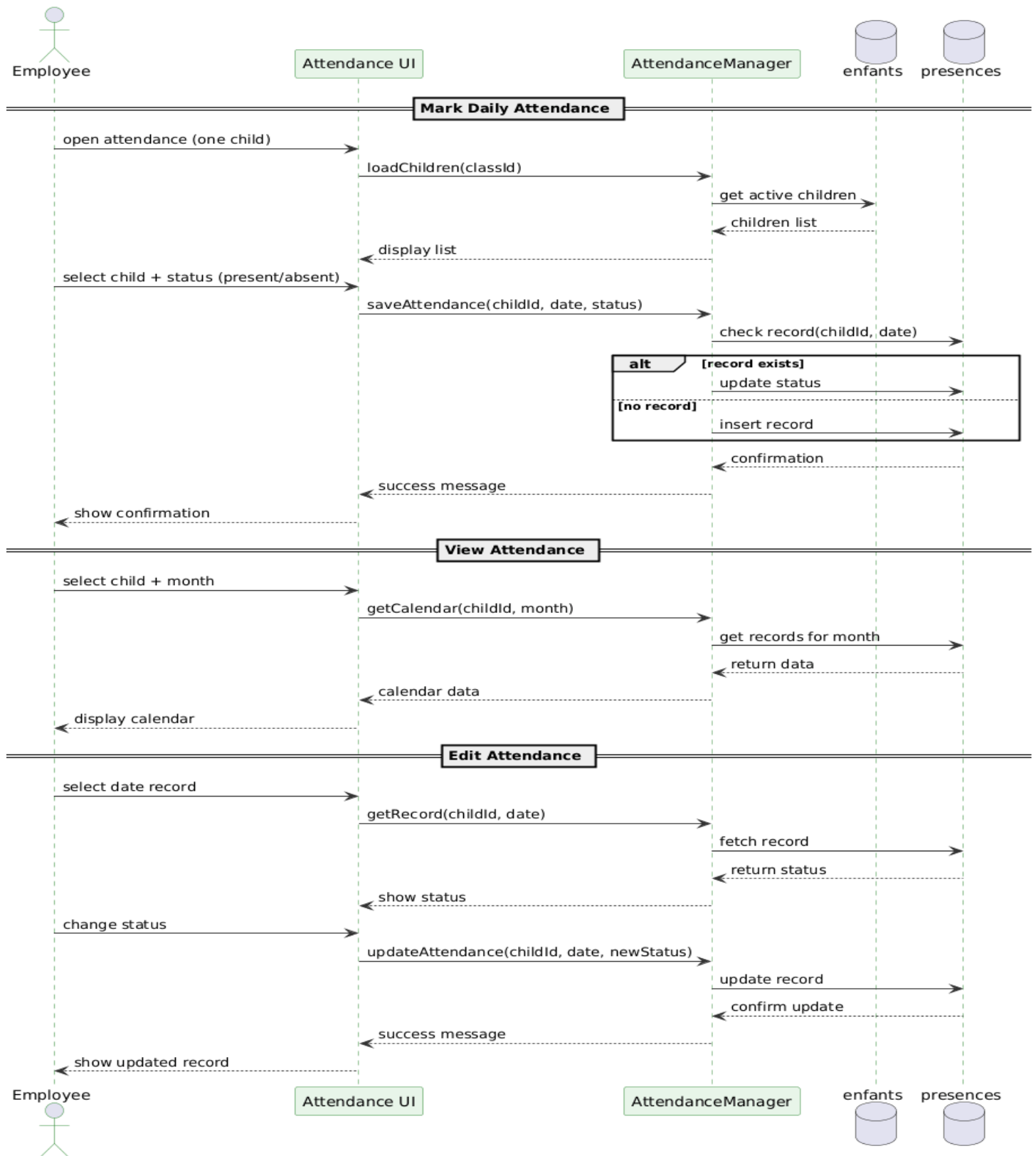


Figure 2.2: Authentication Sequence Diagram

Employee Manaqment:

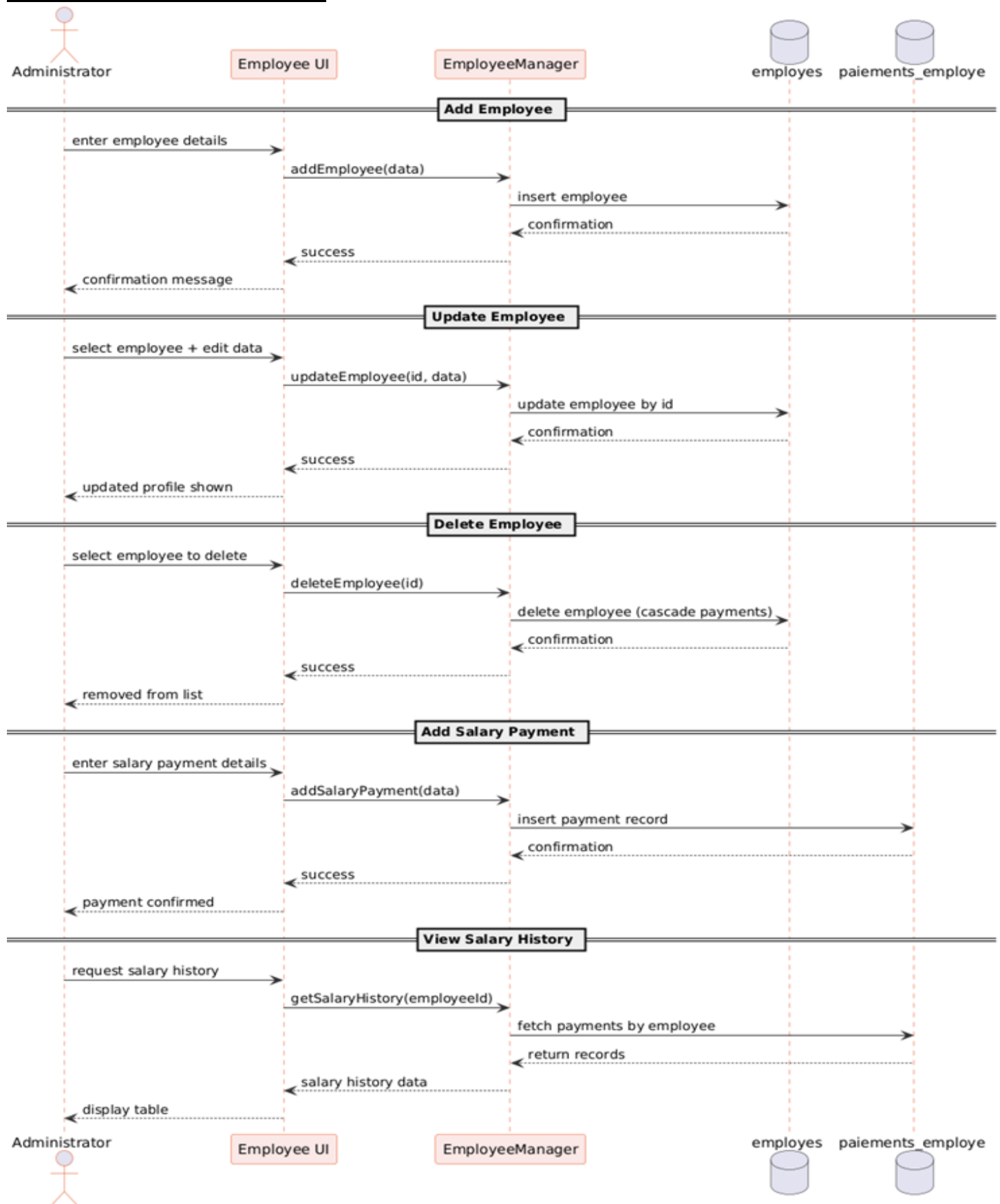


Figure 2.2: Authentication Sequence Diagram

Payment Manaqment:

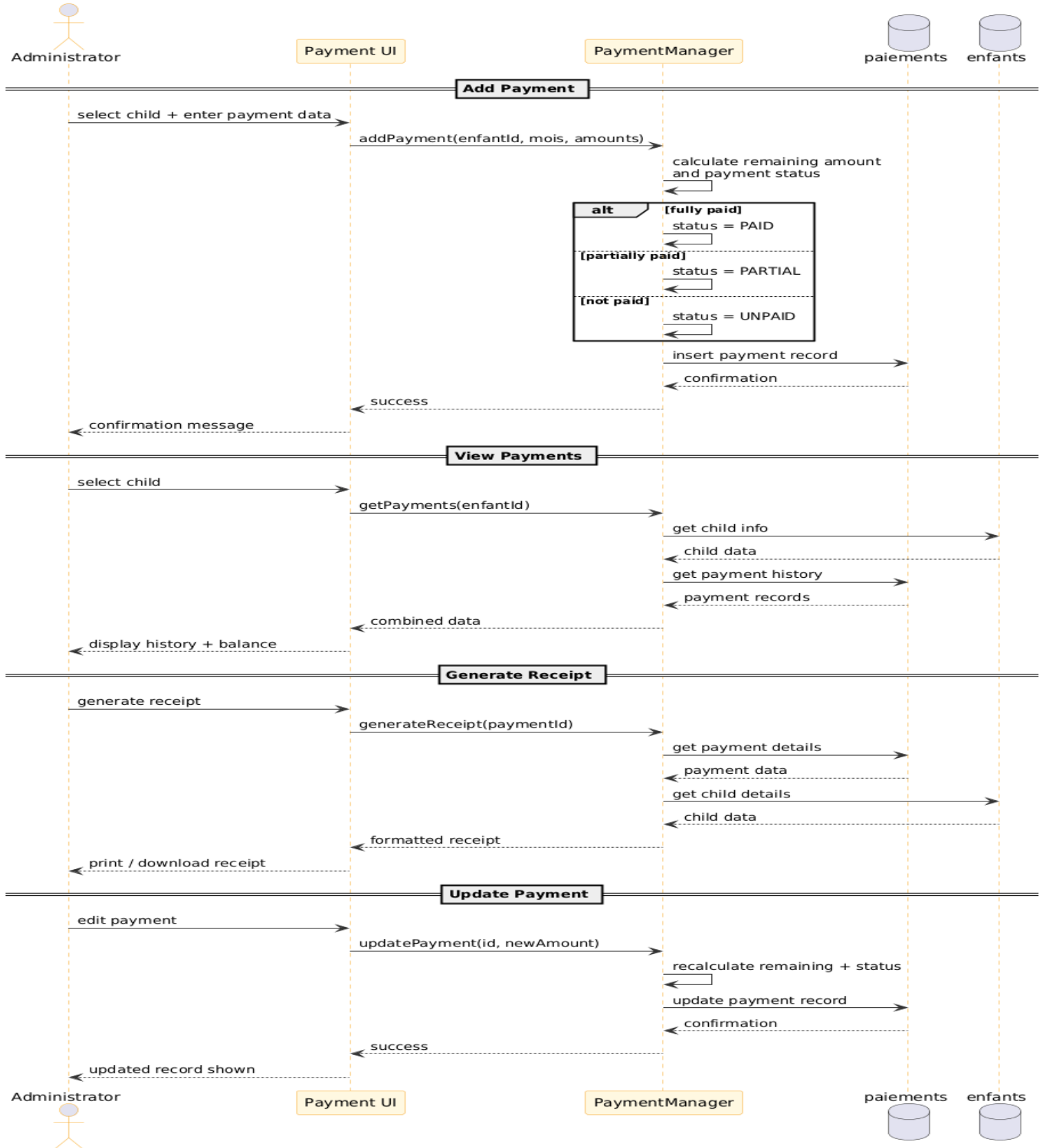


Figure 2.2: Authentication Sequence Diagram

Statistics & Expenses:

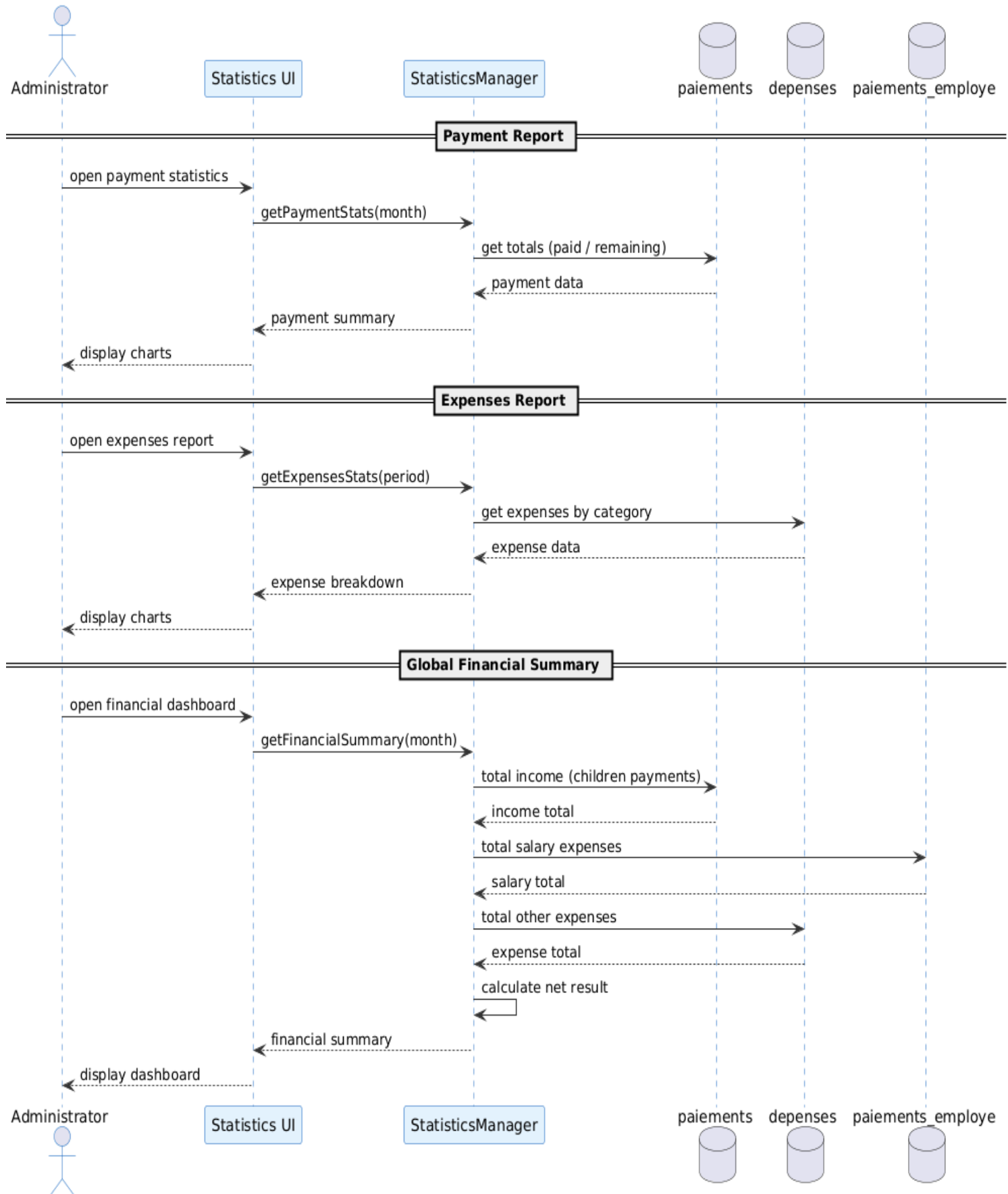


Figure 2.2: Authentication Sequence Diagram

Conclusion

Analysis and design is the topic of discussion in this chapter, where we introduced the reader to UML and the Waterfall approach that was used in developing the system.

Functional analysis of the system followed next, where we established the main actors of the system, their functions, and interactions with each other. In this way, it became easier for us to analyze the requirements and services required of the application.

The design phase followed next, where we modelled the system through UML diagrams. These include global use-case diagrams, class diagrams, database schema, and sequence diagrams. Each of these diagrams helped us represent the system in a structured manner.

Finally, this chapter has laid a solid foundation for the design of the application.

Chapitre 04

Réalisation

Introduction

This chapter describes the implementation process of the nursery management system. It discusses how the implementation of the software is done based on the analysis and design processes discussed in the previous chapter.

At this point, the description of the development environment will be presented, which will discuss the technology tools used in the development process. In addition to this, the architecture of the system will be discussed as well as the way in which the development process is organized.

Finally, in this chapter, the major features of the system will be highlighted using the interface description of the software.

Environnement de développement

Framework Flutter :

Flutter is a software development kit (SDK) provided by Google as an open source tool to develop interactive user interfaces for today's applications. Flutter facilitates the development of powerful apps using one code base.

The use of Flutter in the current project is attributed to the ability to develop appealing interfaces along with easy desktop app development. Furthermore, it provides various pre-made components, which make the development process easy and effective.

Dart Language:

The Dart programming language is employed by Flutter for application development purposes. It is a modern programming language that supports object-oriented features and is easy to learn and use.

In the current project, the programming language Dart is applied to write code for implementation of application logic, handling of data and interaction between UI and database.

Database (SQFLite):

SQFLite refers to a database that enables data storage within the application. This database works on the principle of SQLite and it is widely applied in Flutter apps when it comes to the management of data.

In this project, SQFLite is employed for data storage of everything required for the app, including children's information, attendance records, employees' information, and financial information.

Development Tools (VS Code):

Development Environment

This project has been developed using Visual Studio Code (VS Code). This excellent lightweight code editor supports numerous programming languages and frameworks.

VS Code has been utilized for coding, debugging, and modifying the source code in this project. In addition, it contains some useful plugins for the development of Flutter and Dart applications.



Figure 2.2: Authentication Sequence Diagram

Architecture of the Application

The architecture of the application will indicate the way in which the various modules within the system will be structured and will interact with each other. For the current project, a layered architecture based on the MVC paradigm has been used in order to achieve proper structuring of the application.

In order to organize the application effectively, there are certain layers, such as the user interface, the data models, the database layer, and the logic layer. The user interface module will provide the necessary display and interaction with the user, whereas the data models will consist of the actual structure of the data. The database layer will be concerned with storing and retrieving data, whereas the logic layer will control the data operations.

Project Organization

The project is organized into several main folders, each having a specific role in the application.

Screens:

The *screens* folder contains all the user interfaces of the application. It includes pages such as:

- **login.dart** for user authentication
- **dashboard.dart** for the main overview
- **enfant_screen.dart** for managing children
- **enfant_profil_screen.dart** for displaying detailed child profiles
- **employe_page.dart** and **employe_profile.dart** for employee management
- **présence.dart** and **presence_hist.dart** for attendance tracking and history
- **planing_act.dart** for activity planning
- **paiements_hist.dart** and **suivipaie.dart** for payment tracking
- **depenses.dart** for expense management
- **stats.dart** and **progression.dart** for statistics and reports
- **settings.dart** for application settings

These screens represent the main interaction between the user and the system.

Layout:

The *layout* folder manages the overall structure of the application interface. It includes:

- **main_layout.dart** which defines the main structure of the application
- **sidebar.dart** which provides navigation between different sections

This ensures a consistent design across all screens.

Models:

The *models* folder defines the data structures used in the application. It includes:

- **enfant.dart** for children data
 - **employe.dart** for employees
 - **Presence.dart** for attendance
 - **activity.dart** for activities
 - **depenses.dart** for expenses
 - **accompagneur.dart** for authorized persons
 - **utilisateurs.dart** for user accounts
- These models represent the core entities of the system.

.Database (db):

The *db* folder contains all database-related files and DAO classes. It includes:

- **database_helper.dart** for database configuration and connection
 - DAO files such as **enfant_dao.dart**, **employedaο.dart**, **presencedao.dart**, **activitydao.dart**, **depensesdao.dart**, and others
- These classes handle data storage, retrieval, and manipulation operations.

Services:

The *services* folder contains classes responsible for specific functionalities and logic, such as:

- **statsser.dart** for statistical calculations
 - **ficheenfant.dart** for handling child-related processing
- These services help separate some logic from the interface.

Core Files:

Other important files include:

- **main.dart**, which is the entry point of the application
- **app_state.dart**, which manages the global state of the application
- **trial.service.dart**, which may be used for testing or additional logic

This organization helps keep the project well-structured, making it easier to navigate, understand, and maintain.

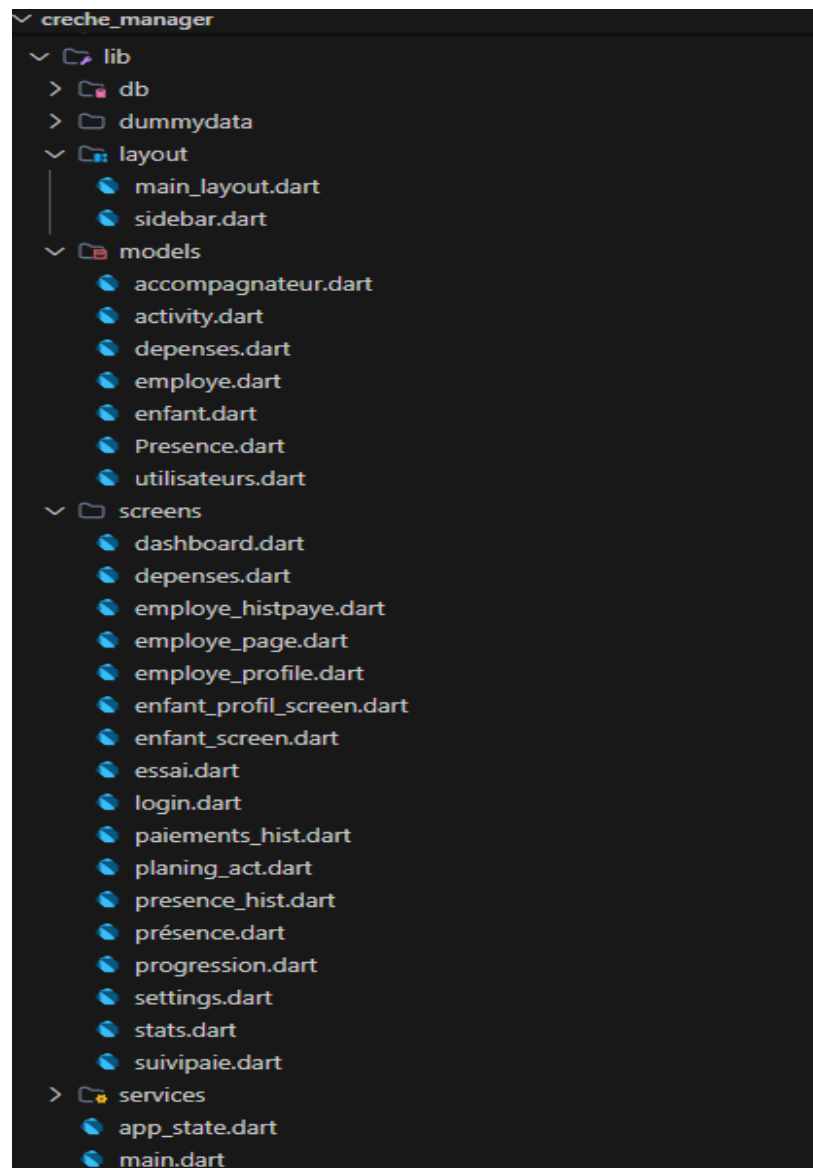


Figure 2.2: Authentication Sequence Diagram

Authentication Interface

CONTENT DE VOUS REVOIR !

Veuillez entrer vos informations

Se connecter



Figure 2.2: Authentication Sequence Diagram

-The dashboard provides the administrator with an overview of the daycare's activity. It displays key information such as the total number of children, monthly registrations, attendance summary, and payment status. It allows the administrator to quickly monitor the overall situation and access different sections of the application.

Dashboard

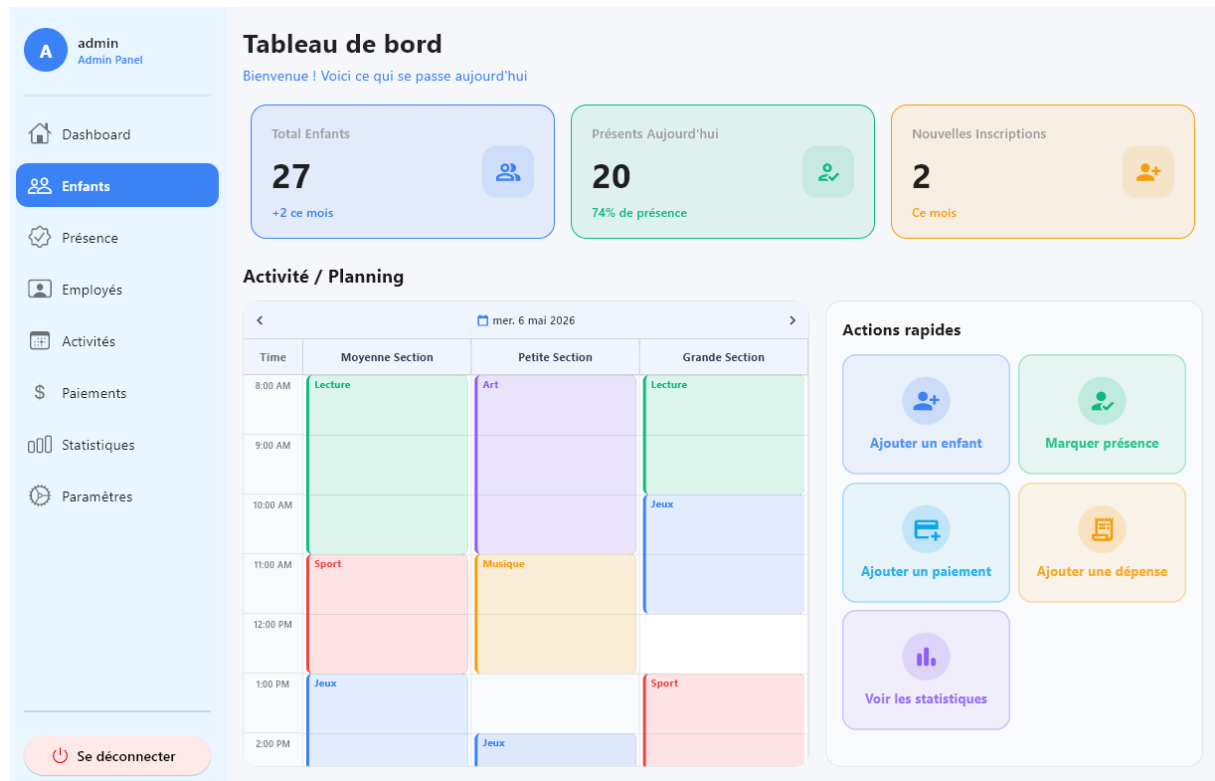


Figure 2.2: Authentication Sequence Diagram

-The authentication interface allows users to access the application by entering their login credentials. The system supports two types of users: **Administrator** and **Standard user**, each with different access levels.

It ensures that only authorized users can enter the system and access its functionalities securely.

Children Management

Tous les enfants

Rechercher Filtrer par Statut Tout + Ajouter un enfant

ID	Enfant	Classe	Téléphone	Statut	Actions
ENF-021	Aliouane Sami	Moyenne Section	0660321987	Actif	
ENF-008	Belkhir Sarah	Moyenne Section	0667890123	Actif	
ENF-001	Benali Yacine	Petite Section	0660123456	Actif	
ENF-022	Bencheikh Layla	Grande Section	0661432098	Actif	
ENF-017	Berkane Wassim	Petite Section	0666543210	Actif	
ENF-005	Boudiaf Adam	Grande Section	0664567890	Actif	
ENF-024	Boudjellal Norah	Petite Section	0663654210	Actif	
ENF-010	Boukhalfa Nour	Petite Section	0669012345	Actif	
ENF-014	Brahimi Ranya	Petite Section	0663210987	Actif	

Se déconnecter < 1 / 3 >

Figure 2.2: Authentication Sequence Diagram

-This screen allows the administrator to manage all children enrolled in the daycare. The administrator can add new children, update their information, delete records, and search or filter the list. It also provides a quick view of each child's basic information and payment status.

Child Profile

Profil de Aliouane Sami

Infos Parentales

Père	Mère
Nom & Prénom Réda Aliouane	Nom & Prénom Aïcha Mahmoudi
Téléphone 0660321987	Téléphone
Profession Technicien	Profession Enseignante
Statut familial Mariés	

Accompagnateurs autorisés

+ Ajouter un accompagnateur

Infos Santé

Allergies	Commentaires médicaux
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Actions:

- DÉSACTIVER ENFANT
- IMPRIMER FICHE ENFANT
- IMPRIMER CARTE ENFANT
- HISTORIQUE DE PRÉSENCE
- HISTORIQUE DES PAIEMENTS
- EVALUATION

Se déconnecter

Date d'inscription : 26/03/2021

Figure 2.2: Authentication Sequence Diagram

-The child profile screen provides detailed information about a selected child. The administrator can view and update personal data, health information, parents' details, and authorized pickup persons. It also allows access to attendance history and payment records, as well as printing the child's file and ID card.

Attendance History

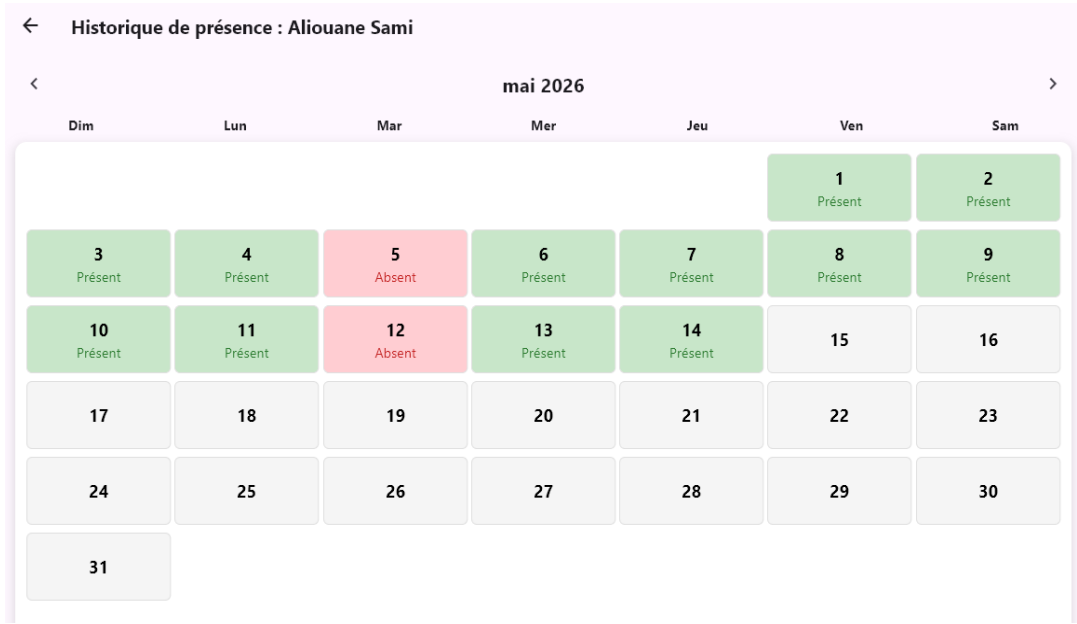


Figure 2.2: Authentication Sequence Diagram

Payments History

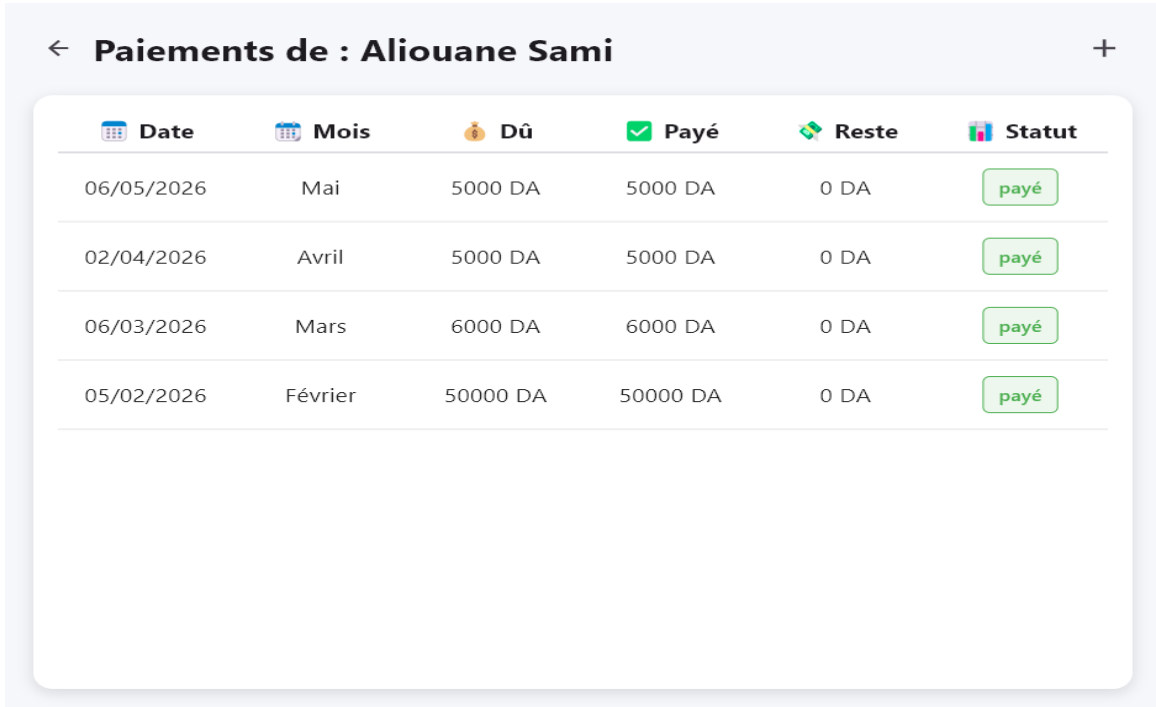


Figure 2.2: Authentication Sequence Diagram

Evaluations

← Evaluation de Sami Aliouane		← Mai 2026 →	
Respecte les règles de vie, de la classe et de l'école	☐	☐	
S'organise et travaille seul(e)	☐	☐	
Désire apprendre et manifeste de la curiosité	☐	☐	
Comprend et respecte les consignes	☐	☐	
Mène un travail à son terme	☐	☐	
Retient des comptines, des poèmes	☐	☐	
Écoute la maîtresse	☐	☐	
2) Compétence langue française			
Prend la parole et argumente (dans le sujet)	☐	☐	
Utilise des phrases assez élaborées	☐	☐	
Utilise un vocabulaire précis	☐	☐	
Reconnaît les lettres	☐	☐	
Reconnaît certains mots dans une phrase	☐	☐	
Repère des correspondances lettres/sons	☐	☐	
Utilise correctement crayon, stylo, craie	☐	☐	
Écrit seul(e) des lettres	☐	☐	
Écrit correctement entre deux lignes	☐	☐	
3) Mathématiques			
Connaît la suite des nombres jusqu'à	☐	☐	
Classe les nombres	☐	☐	
Écrit correctement les chiffres	☐	☐	
Sait compter un certain nombre d'objets	☐	☐	
Reproduit et nomme des formes simples	☐	☐	
Compare des grandeurs	☐	☐	

Figure 2.2: Authentication Sequence Diagram

Attendance Managment

Élève	Classe	Statut
Sami Aliouane	Moyenne Section	✗ Absent
Sarah Belkhir	Moyenne Section	✓ Présent
Inès Cherif	Moyenne Section	✓ Présent
Adel Ferhat	Moyenne Section	✓ Présent
Yasmine Larbi	Moyenne Section	✗ Absent
Amine Mansouri	Moyenne Section	✓ Présent
Sara	Moyenne Section	✓ Présent
Zakaria Slimani	Moyenne Section	✓ Présent
Selma Touati	Moyenne Section	✓ Présent

Figure 2.2: Authentication Sequence Diagram

-This screen allows the administrator to track and manage children's attendance. The administrator can mark children as present or absent and view attendance history in a calendar format. It helps in monitoring daily presence and ensuring accurate records.

Employee Managment

The screenshot displays the 'Employés' management interface. On the left is a sidebar with a user profile 'admin Admin Panel' and navigation links: Dashboard, Enfants, Présence, **Employés**, Activités, Paiements, Statistiques, and Paramètres. At the bottom of the sidebar is a 'Se déconnecter' button. The main area is titled 'Employés' and includes a search bar, a 'Filtrer par' dropdown, a 'Statut' dropdown set to 'Tout', and a 'Employés par page' dropdown set to '5'. A green '+ Ajouter un employé' button is located below the filters. The table below lists four employees with columns for ID, Employé, Poste, Téléphone, Statut, and Actions.

ID	Employé	Poste	Téléphone	Statut	Actions
EMP-001	Karima Benhamou	Directeur	0550100200	Actif	
EMP-002	Sabrina Hamoudi	Éducatrice	0660200300	Actif	
EMP-003	Nassima Lakhdari	Aide-éducatrice	0770300400	Actif	
EMP-004	Mohamed Bouhadda	Agent d'entretien	0550400500	Actif	

< 1 / 1 >

Figure 2.2: Authentication Sequence Diagram

- The employee management screen enables the administrator to manage staff information. The administrator can add new employees, update their details, delete records, and view their profiles. This helps in organizing the daycare staff efficiently

Activities Managment

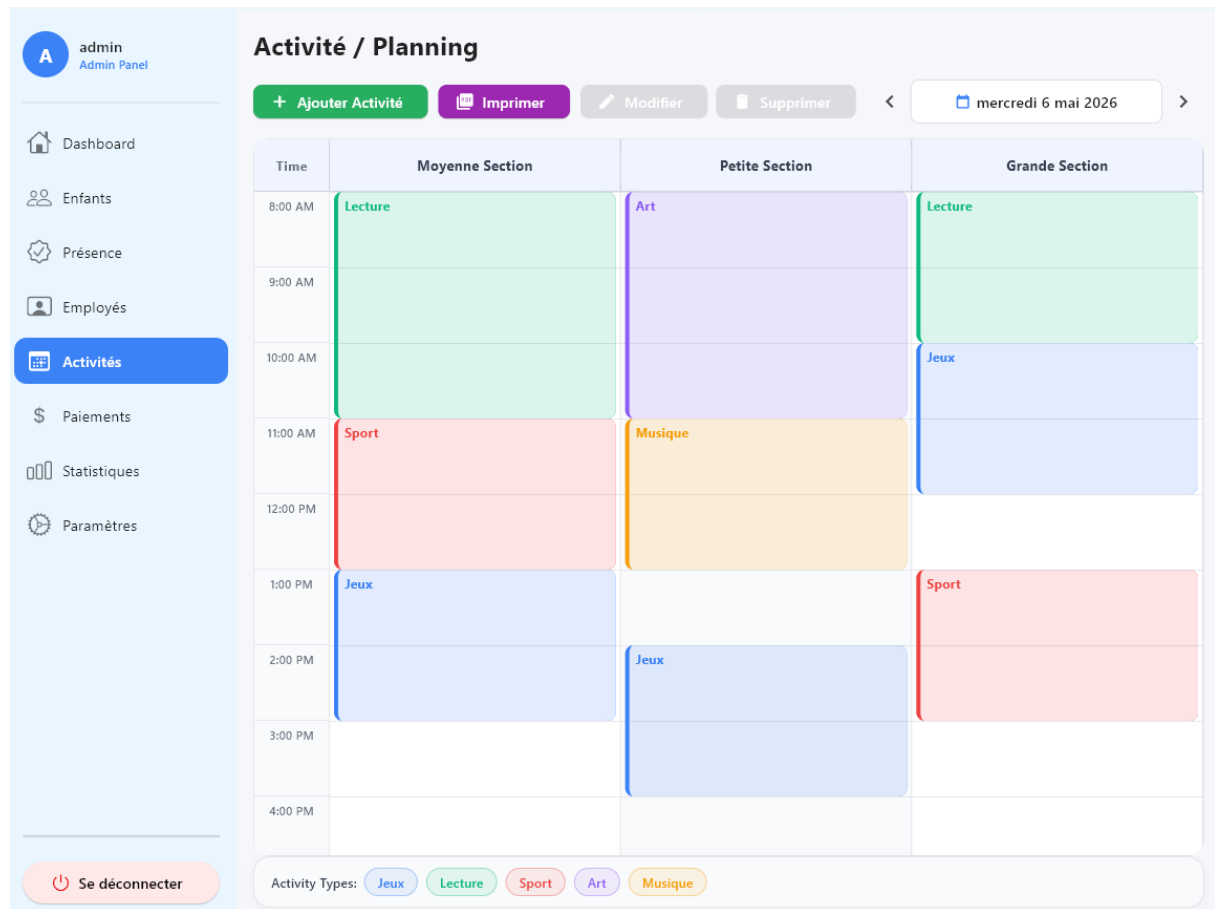


Figure 2.2: Authentication Sequence Diagram

-This screen allows the administrator to organize and manage children's activities and planning. The administrator can create schedules, assign activities, and ensure proper organization of daily programs.

Expenses Managment

AadminAdmin Panel Dashboard Enfants Présence Employés Activités Paielements Dépenses Paielements Enfant Statistiques Paramètres Se déconnecter	Dépenses				
	Filtre actif : Aucun				
	Categorie	Description	Montant	Date	Attestation
	Électricité	Facture Sonelgaz — Avril 2026	18500.00 DA	03/05/2026	
	Alimentation	Achat de fruits et légumes pour la semaine	8500.00 DA	02/05/2026	
	Salaires	Salaire de Karima Benhamou - 2026-04	105000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Karima Benhamou - 2026-04	105000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Karima Benhamou - 2026-04	105000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Sabrina Hamoudi - 2026-04	55000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Sabrina Hamoudi - 2026-04	55000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Sabrina Hamoudi - 2026-04	55000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Nassima Lakhdari - 2026-04	45000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Nassima Lakhdari - 2026-04	45000.00 DA	30/04/2026	Paielement employé
	Salaires	Salaire de Nassima Lakhdari - 2026-04	45000.00 DA	30/04/2026	Paielement employé
	Fournitures	Cahiers, crayons et feutres	12000.00 DA	28/04/2026	
	Entretien	Produits de nettoyage et désinfectants	6500.00 DA	25/04/2026	
	Maintenance	Réparation climatiseur salle Moyenne Section	14000.00 DA	20/04/2026	

Figure 2.2: Authentication Sequence Diagram

- This screen allows the administrator to manage the daycare's expenses. The administrator can add, update, and track different types of expenses, helping to maintain better financial control.

Payments Mangament

Suivi des paiements						
Rechercher... Classe Mois Statut + Ajouter Paiement						
Enfant	Mois concerné	Montant dû	Montant payé	Solde restant	Date paiement	Statut
Aliouane Sami	Mai	5000 DA	5000 DA	0 DA	06/05/26	✓ Payé
Aliouane Sami	Avril	5000 DA	5000 DA	0 DA	02/04/26	✓ Payé
Aliouane Sami	Mars	6000 DA	6000 DA	0 DA	06/03/26	✓ Payé
Aliouane Sami	Février	50000 DA	50000 DA	0 DA	05/02/26	✓ Payé
Belkhir Sarah	2026-05	12000 DA	12000 DA	0 DA	05/05/26	✓ Payé
Belkhir Sarah	2026-04	12000 DA	6000 DA	6000 DA	05/04/26	⚠ Partiel
Benali Yacine	2026-05	12000 DA	12000 DA	0 DA	05/05/26	✓ Payé
Benali Yacine	2026-04	12000 DA	12000 DA	0 DA	05/04/26	✓ Payé
Boudiaf Adam	2026-05	12000 DA	6000 DA	6000 DA	05/05/26	⚠ Partiel
Boudiaf Adam	2026-04	12000 DA	12000 DA	0 DA	05/04/26	✓ Payé
1 – 10 sur 24						

Figure 2.2: Authentication Sequence Diagram

- The payment management screen allows the administrator to handle all financial transactions related to children. The administrator can record payments, track payment status (paid, unpaid, partial), and view payment history. It also supports generating payment receipts.

Statistics and Charts

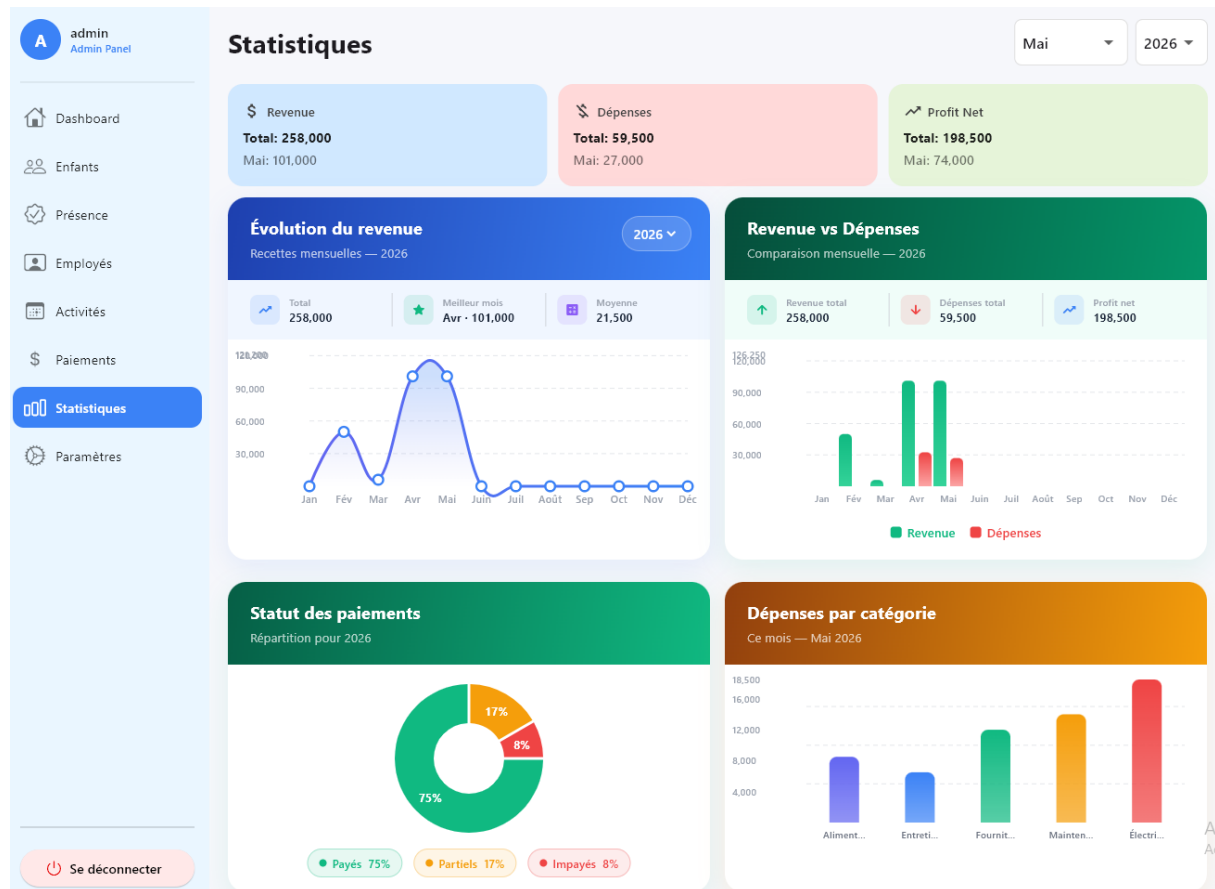


Figure 2.2: Authentication Sequence Diagram

- The statistics screen provides visual reports about the daycare's activity. The administrator can view charts related to children, attendance, and payments, which helps in analyzing performance and making better decisions.

Settings

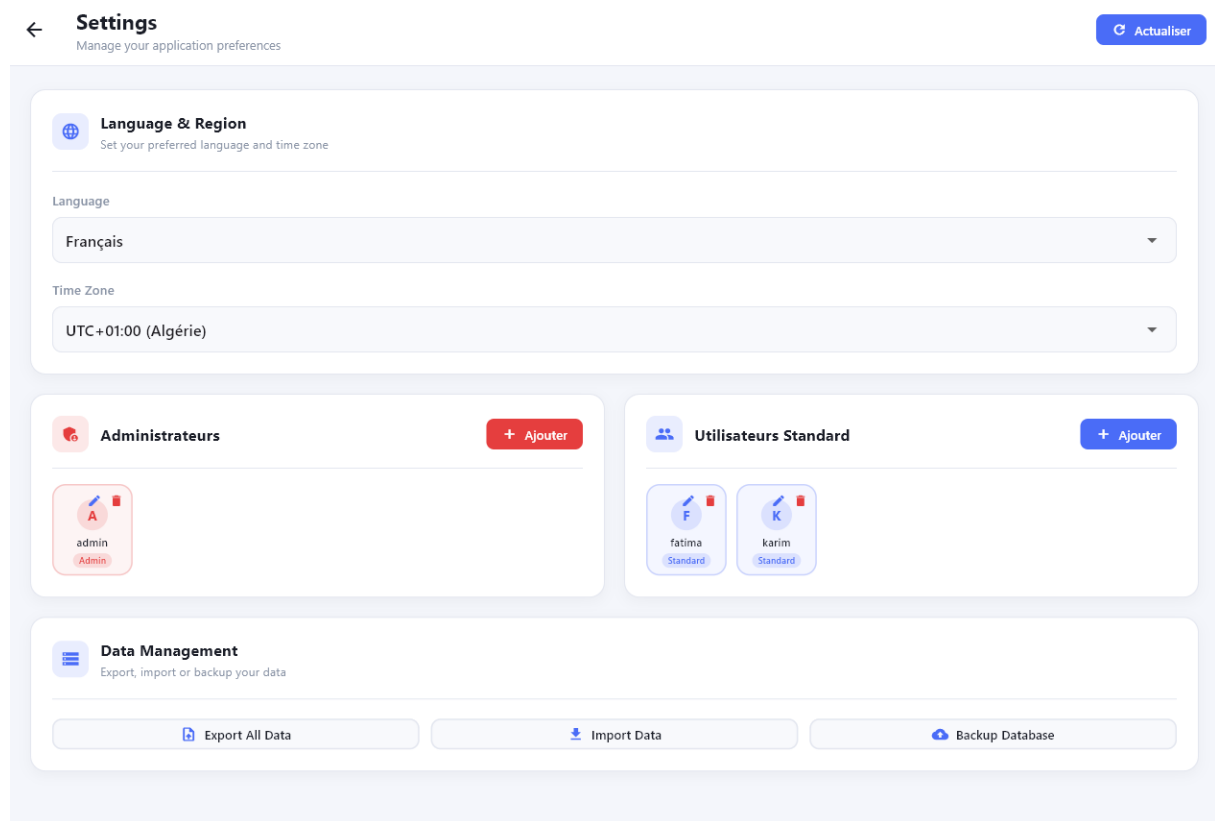


Figure 2.2: Authentication Sequence Diagram

-The settings screen allows the administrator to configure the application. It may include options for managing user accounts, updating preferences, and controlling general system settings

General Conclusion

From this project, we developed a nursery management app that is intended to enhance and simplify the day-to-day operations of daycare centers. The primary goal of this study is to eliminate the conventional way of managing the facility using traditional means like paperwork and other primitive techniques by introducing a computerized method.

In this project, we researched the existing system and found out the main challenges experienced when running a daycare center, namely information management, monitoring attendance, controlling the workforce, and following up on payments. Based on this information, we formulated and designed a desktop application tailored to this purpose.

Our app comprises numerous functions, such as management of children, attendance management, control over employees, scheduling activities, monitoring payments and expenses, as well as creating reports. These functions allow for better organization, decrease in manual labor, and easy access to data.

This project was also an opportunity for us to practice our programming skills through the use of Flutter, Dart, and SQFLite programming languages and UML modeling.

Lastly, our application could still be enhanced in the future by incorporating additional functions such as interaction with parents, online services, notifications, and cloud backup.